Data and Annotation Needs

1. Purpose

This file lists the data and annotation details needed for future research. It is intended to reduce ambiguity before collecting or labeling new samples.

2. Required Data Types

2.1 Handwriting

Needed:

• scanned or photographed handwriting pages
• tablet or stylus handwriting samples
• optionally stroke-order or pen-dynamics data

Useful labels:

• learning-disorder / not flagged
• mild / moderate / severe
• dysgraphia-related handwriting difficulty

2.2 Reading audio

Needed:

• spoken reading samples
• prompt text for each recording
• metadata about language and reading context

Useful labels:

• pronunciation errors
• hesitations
• repetitions
• omissions
• reading fluency level

2.3 Text samples

Needed:

• short reading passages
• sentence reading prompts
• spelling task inputs

Useful labels:

• error count
• error type
• language tag

2.4 Reading behavior

Needed:

• reading time
• hesitation count
• repetition count
• omission count

Useful labels:

• fluency level
• task difficulty
• progress over time

2.5 Eye-tracking and visual focus

Needed:

• gaze traces
• fixation and saccade records
• visual-focus test outcomes

Useful labels:

• reading speed
• regressions
• fixation duration
• attention stability

2.6 Biomarker tables

Needed:

- numeric feature tables
- clear label columns
- consistent sample identifiers

Useful labels:

- screening label
- severity label
- language group

3. Annotation Guidelines

3.1 Consistency

- use the same label definitions across all annotators
- keep language tags explicit
- keep severity scales documented

3.2 Anonymization

- do not store names in manifests
- do not store phone numbers or school IDs unless absolutely required and consented
- prefer stable hashed identifiers

3.3 Missing values

- keep missing modality indicators explicit
- do not silently invent values
- record whether a missing value means not collected or not applicable

4. Recommended Metadata

Useful extra metadata fields:

- age group
- grade
- language

• school region
• device type
• annotator ID
• collection date

5. Data Quality Checks

• verify required columns exist
• check for duplicate sample IDs
• verify file paths resolve correctly
• inspect class imbalance
• inspect language coverage
• confirm audio duration is non-zero
• confirm handwriting files are readable

6. Research Implications

Higher-quality annotation will help most with:

• cross-lingual transfer
• severity modeling
• biomarker ranking
• intervention evaluation
• explainability validation
• trustworthy cross-validation selection and holdout evaluation